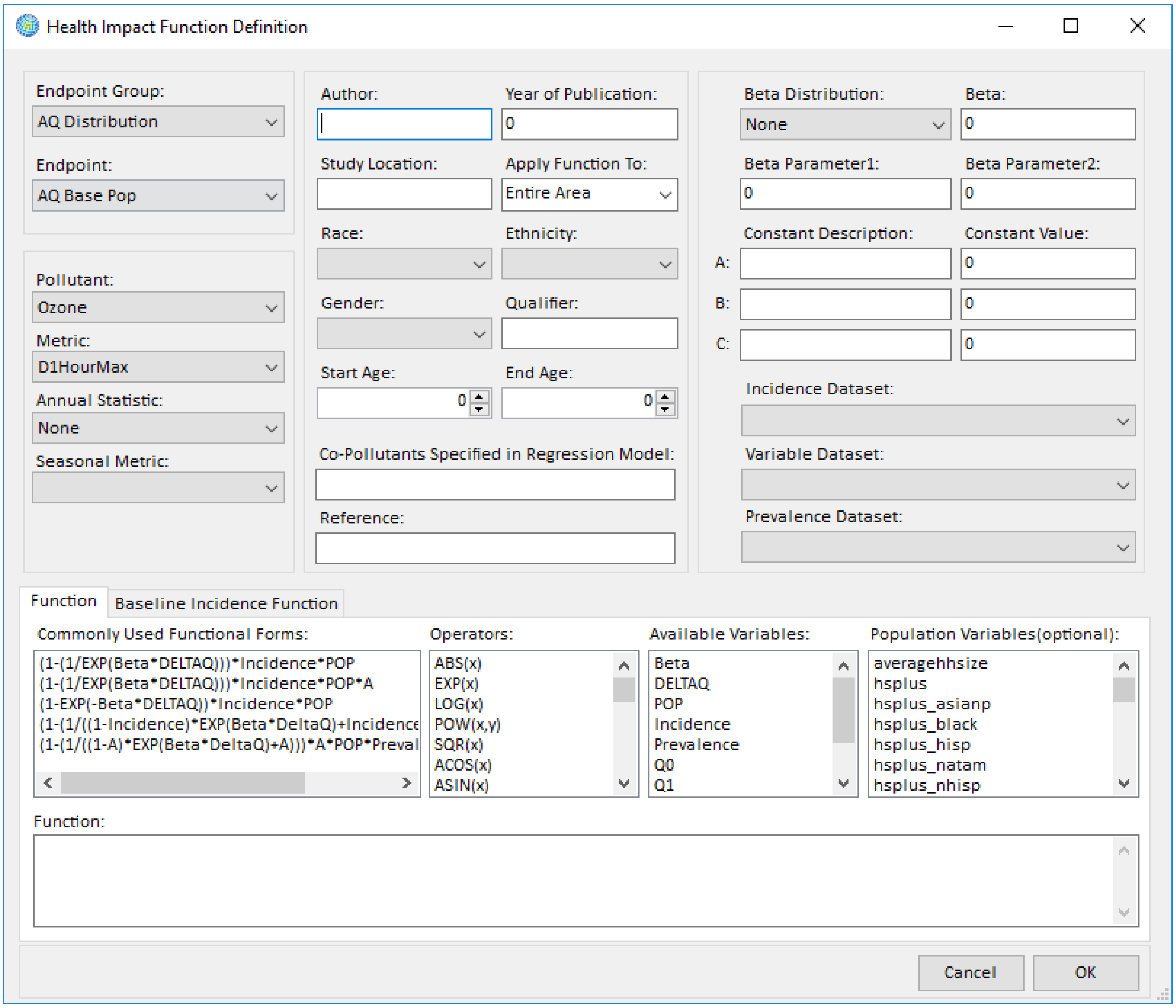

From the **Health Impact Function Dataset Definition** window you can also add health impact functions to the ones that are already in your dataset. Click the **Add** button and a blank **Health Impact Function Definition** window will appear and you can then create new health impact functions. (See Appendix M: Function Editor for additional information about the syntax for developing functions with this editor.)

Starting in Version 1.4, you can link a health impact function to a specific geographic area for which you have uploaded a grid definition by selecting that area in the **Apply Function To** dropdown menu. This prevents you from applying the function to other grid definitions but may be appropriate for functions derived using sub-national data. By default this menu is set to “Entire Area”, which means the application of the health impact function is unrestricted. Note that if an air quality cell intersects multiple geographic areas, BenMAP-CE will calculate health impacts for the geographic area containing the majority of the air quality cell.

After defining the new health impact function, click **OK**. This will take you back to the **Health Impact Function Dataset Definition** window. When you are finished with any editing or adding of health impact functions, click **OK.** From the **Manage Health Impact Function Datasets** window, you can also select an **Available Dataset** and **Data row** and view the Metadata. To view the Metadata associated with the data file, click the **View Metadata** button to view and edit existing references and descriptions. Click **OK** on the **Manage Health Impact Function Datasets** window when you are satisfied with all your inputs. The **Modify Datasets** window will appear. Here in the **Health Impact Functions** box you should see an entry for any health impact function datasets that you have loaded.

4.1.6.2 Format for Health Impact Functions

Table 4-9 presents the variables that can be used in health impact function datasets.

Table 4-9. Health Impact Function Dataset Variables

Variable	Type	Required	Notes
Endpoint Group	Text	Yes	If this does not reference an already defined Endpoint Group, one will be added.
Endpoint	Text	Yes	If this does not reference an already defined Endpoint for the Endpoint Group, one will be added.
Pollutant	Text	Yes	Should reference an already defined Pollutant.
Metric	Text	Yes	Should reference an already defined Metric for the Pollutant.
Annual Statistic	Text	No	Should either be blank (signifying no annual metric value) or be one of: None, Mean, Median, Min, Max, Sum.
Seasonal Metric	Text	No	Should either be blank (signifying no Seasonal Metric value) or reference an already defined Seasonal Metric for the Metric.
Race		No	Should either be blank (signifying All Races) or reference a defined Race.
Ethnicity		No	Should either be blank (signifying All Ethnicities) or reference a defined Ethnicity.
Gender		No	Should either be blank (signifying All Genders) or reference a defined Gender.
Start Age	Integer	Yes	Specifies the low and high ages, inclusive. For example, Start Age of '*0*' and End Age of '*1*' includes infants through the first 12 months of life and all one-year old infants.
End Age	Integer	Yes	
Author	Text	No	The author(s) of the study from which the function is derived.
Apply Function To		No	The specific geographic area to which you would like to apply the health impact function; unrestricted by default ("Entire Area")

Variable	Type	Required	Notes
Year of Publication	Integer	Yes	The year of publication of the study.
Qualifier	Text	No	Provides additional information to identify a particular health impact function, such as when a particular study has multiple functions.
Location Name		No	Type of study area. For the 'United States' setup, choose between 'State', 'County', and 'MSA (metropolitan area)'.
Location	Text	No	The specific location of the study.
Co-Pollutants Specified in Regression Model	Text	No	Identifies other pollutants that were included simultaneously in the estimation equation for the pollutant of interest.
Reference	Text	No	Bibliographic reference, included to identify the source in the health literature.
Function	Text	Yes	The functional form, interpreted (executed) by BenMAP-CE when running an analysis to estimate air pollution-related health impacts. For example, the log-linear form is as follows: '*(1-(1/ EXP(Beta*DELTAQ)))*Incidence*POP*'.
Baseline Incidence Function	Text	Yes	The functional form, interpreted (executed) by BenMAP-CE to estimate health impacts due to all causes. This typically has the form: '*Incidence*POP*'.
Beta Distribution	Text	No	If the Beta has no distribution, any value is acceptable. Otherwise, should be one of: *Normal*, *Triangular*, *Poisson*, *Binomial*, *LogNormal*, *Uniform*, *Exponential*, *Geometric*, *Weibull*, *Gamma*, *Logistic*, *Beta*, *Pareto*, *Cauchy*, *Custom*.
Beta	Numeric (double)	No	Mean value of the Beta distribution.
Beta Parameter 1	Numeric (double)	No	Parameter 1 of the Beta distribution (meaning depends on the distribution - for Normal distributions this represents the standard deviation).
Beta Parameter 2	Numeric (double	No	Parameter 2 of the Beta distribution (meaning depends on the distribution - for Normal distributions this is not required).
Name A	Text	No	Description of variable A.
A	Numeric (double)	No	A constant value which can be referenced by the Function.
Name B	Text	No	Description of variable B.
B	Numeric (double)	No	A constant value which can be referenced by the Function.

Variable	Type	Required	Notes
Name C	Text	No	Description of variable C.
C	Numeric (double)	No	A constant value which can be referenced by the Function.
Incidence Dataset	Text	No	Specifies the dataset from which incidence data will be derived. The user may choose from multiple datasets. (Initially this field may be left blank.) See Section 4.1.4.
Prevalence Dataset	Text	No	Specifies the dataset from which prevalence data will be derived. The user may choose from multiple datasets. (Initially this field may be left blank.) See Section 4.1.4.
Variable Dataset	Text	No	Specifies the dataset from which "variable" data (e.g., income data) will be derived. The user may choose from multiple datasets. (Initially this field may be left blank.) See Section 4.1.7.

While the mean value of the Beta distribution is always available in BenMAP-CE, the values associated with Beta Parameter 1 and Beta Parameter 2 may change depending on the type of distribution. Table 4-10 provides a list of the variables associated with each type of beta distribution in BenMAP-CE.

Table 4-10. Beta Distribution Types and Variables

Distribution	Formula	Beta Parameter 1	Beta Parameter 2	Notes
Normal	$\frac{1}{\sigma\sqrt{2\pi}} \times e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	Standard deviation (sigma)	N/A	The Normal distribution has two parameters - the mean, mu, and the standard deviation, sigma.
Triangular	$\frac{2(x-a)}{(b-a)(c-a)}$ for $a \leq x \leq c$ $\frac{2(b-x)}{(b-a)(b-c)}$ for $c \leq x \leq b$	Minimum value (a)	Maximum value (b)	The Triangular distribution has three parameters - the minimum value (a), the maximum value (b), and the most likely value (c). BenMAP-CE uses the mean value, the minimum, and the maximum to calculation the most likely value.
Poisson	$\lambda^x e^{-\lambda} / x!$	Lambda	N/A	The Poisson distribution has a single parameter, lambda.
Binomial	$\frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}$	n	p	The Binomial distribution has two parameters, n and p.
LogNormal	$1/\sigma\sqrt{2\pi} \cdot e^{-(x-\mu)^2/2\sigma^2}$	Standard deviation (sigma) of the corresponding	N/A	The LogNormal distribution has two parameters - the mean of the corresponding Normal distribution, mu, and the